IBM University Engagement Project Submission

“Exploring the Power of Agentic AI with ***IBM Granite***”

Project Tile – Research-Agent

Domain of Project – [Artificial Intelligence/ Information Retrieval & Research Automation]

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Problem Statement –

Problem statement-Research Agent

The Challenge

A Research Agent is an AI system designed to assist with academic and scientific research tasks. It can autonomously search for **literature, summarize papers, and organize references**. Using **natural language processing, it understands research questions and retrieves relevant information**. The agent can generate **reports, suggest hypotheses, and even draft sections of research papers**. It saves time by automating repetitive tasks like **citation management and data extraction**. Research Agents enhance **efficiency, accuracy, and innovation in both academic and industrial R&D**.

Objective

The objective of this project is to build an AI Research Agent that automates **information retrieval, analysis, and summarization, enabling users to obtain accurate research** insights **quickly and efficiently**.

- **Information Retrieval**

Collect relevant information from websites, documents, and databases.

- **Data Analysis**

Analyse gathered information to identify key insights and trends.

- **Research Automation**

Reduce manual effort by automating the research workflow.

- **Improved Decision-Making**

Provide accurate and relevant information to support informed decisions.

Technology Used

- **IBM watsonx.orchestrate** – Large Language Model integration and AI inference.
- **Agentic AI** – Enables autonomous research, reasoning, and response generation.
- **Retrieval-Augmented Generation (RAG)** – Retrieves relevant information before generating answers.
- **Natural Language Processing (NLP)** – Understands and processes user queries.
- **Web Search Integration** – Collects information from online sources.
- **Dataset Processing & Analysis** – Extracts insights from structured data.

Proposed Solution -

The proposed solution is an AI Research Agent developed using **IBM Cloud** and **IBM watsonx Orchestrate**. The system leverages Large Language Models (LLMs), Natural Language Processing (NLP), and Retrieval-Augmented Generation (RAG) to automate the research process.

When a user submits a query, the agent intelligently understands the request, retrieves relevant information from multiple sources, analyses the **collected data, and generates concise summaries and structured reports**. Watsonx Orchestrate manages and coordinates the workflow between different AI services, enabling efficient information retrieval, analysis, and report generation. By automating repetitive research tasks, the solution reduces **manual effort, improves research accuracy, and delivers reliable insights to users in a timely manner.**

The System Consists Of **3 key agents** for **AI Research Agent**

- **Query Understanding Agent**

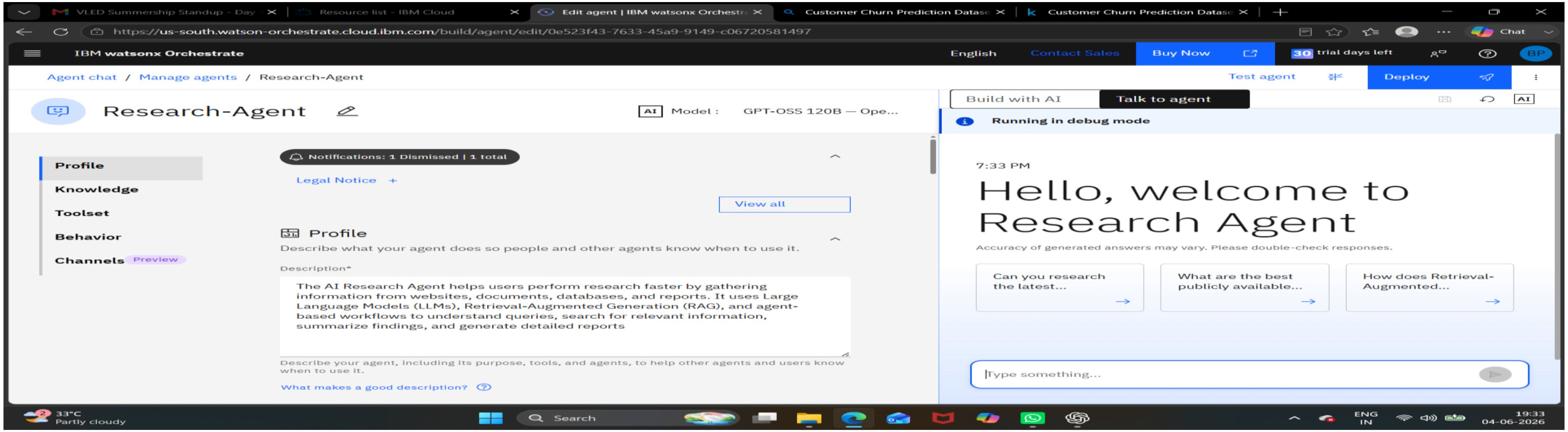
Understands and interprets user research queries using Natural Language Processing (NLP) to identify the user's intent and information requirements.

- **Information Retrieval Agent**

Searches and collects relevant information from websites, documents, databases, and other trusted sources based on the user's query.

- **Report Generation Agent**

Analyses and summarizes the collected information, then generates a structured research report with key findings and insights.


IBM watsonx Orchestrate
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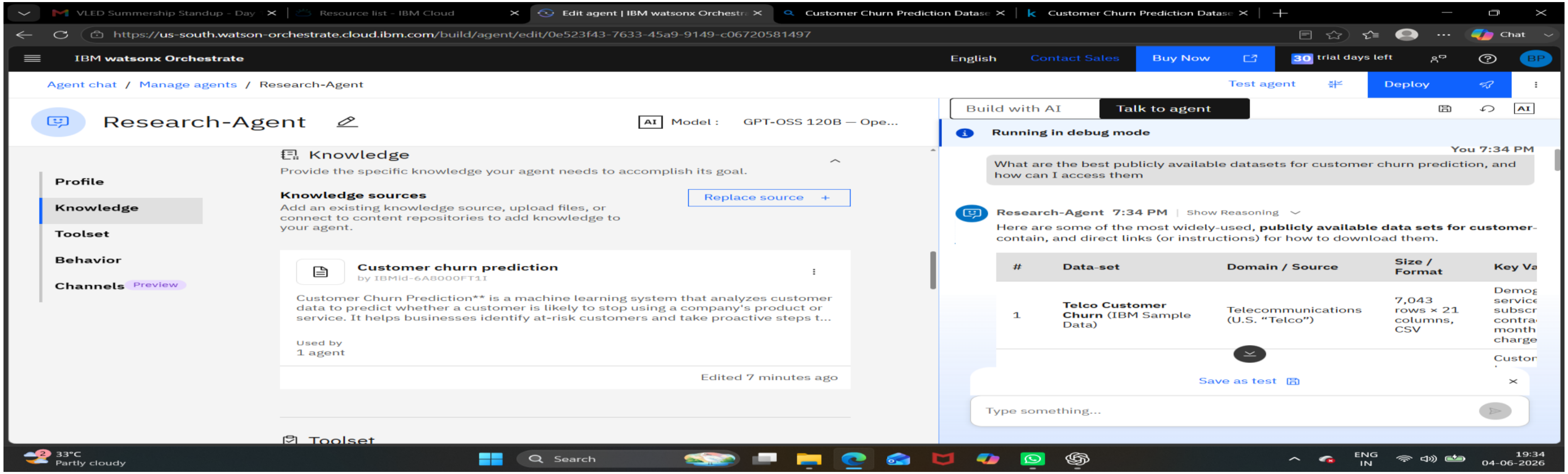
Agent chat / Manage agents / Research-Agent
Test agent
Deploy
Build with AI
Talk to agent
Research-Agent
Model : GPT-OSS 120B — Ope...
Running in debug mode
Profile
Knowledge
Toolset
Behavior
Channels Preview
Notifications: 1 Dismissed | 1 total
Legal Notice +
View all
Profile
Describe what your agent does so people and other agents know when to use it.
Description*
The AI Research Agent helps users perform research faster by gathering information from websites, documents, databases, and reports. It uses Large Language Models (LLMs), Retrieval-Augmented Generation (RAG), and agent-based workflows to understand queries, search for relevant information, summarize findings, and generate detailed reports
Describe your agent, including its purpose, tools, and agents, to help other agents and users know when to use it.
What makes a good description?
7:33 PM
Hello, welcome to Research Agent
Accuracy of generated answers may vary. Please double-check responses.
Can you research the latest...
What are the best publicly available...
How does Retrieval-Augmented...
Type something...


IBM watsonx Orchestrate
English
Contact Sales
Buy Now
30 trial days left
Agent chat / Manage agents / Research-Agent
Test agent
Deploy
Build with AI
Talk to agent
Research-Agent
Model : GPT-OSS 120B — Ope...
Running in debug mode
Knowledge
Provide the specific knowledge your agent needs to accomplish its goal.
Profile
Knowledge
Toolset
Behavior
Channels Preview
Knowledge sources
Add an existing knowledge source, upload files, or connect to content repositories to add knowledge to your agent.
Replace source +
Customer churn prediction
Customer Churn Prediction** is a machine learning system that analyzes customer data to predict whether a customer is likely to stop using a company's product or service. It helps businesses identify at-risk customers and take proactive steps t...
Used by
1 agent
Edited 7 minutes ago
Toolset
You 7:34 PM
What are the best publicly available datasets for customer churn prediction, and how can I access them
Research-Agent 7:34 PM
Show Reasoning
Here are some of the most widely-used, publicly available data sets for customer-contain, and direct links (or instructions) for how to download them.
#
Data-set
Domain / Source
Size / Format
1
Telco Customer Churn (IBM Sample Data)
Telecommunications (U.S. "Telco")
7,043 rows × 21 columns, CSV
Save as test
Type something...

Architecture

USER

UI (Chat Interface)

IBM watsonx Orchestrate
(Agent Workflow & Planning)

watsonx.ai (LLM)

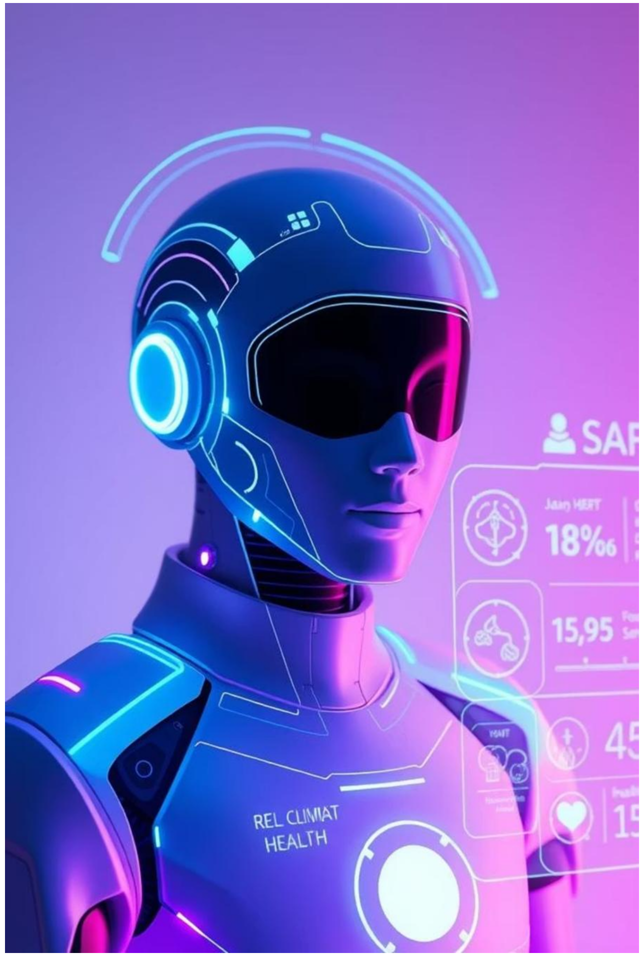
RAG Engine (Retrieve + Generate)

Web Data + Dataset Knowledge Base

Final Answer / Report

Role of Agentic AI in the solution

Agentic AI plays a crucial role in the **AI Research Agent** by enabling the system to act autonomously and perform research tasks with **minimal human intervention**. It understands **user queries, plans the research process, retrieves relevant information** from multiple sources, **analyses** and **summarizes the collected data**, and **generates structured reports**. Through **IBM watsonx Orchestrate**, Agentic AI coordinates different agents and workflows to ensure efficient task execution. This approach reduces **manual effort, improves research accuracy, accelerates information discovery, and provides users** with reliable insights in a timely manner.



Novelty and Uniqueness

The proposed **AI Research Agent** stands out by combining **Agentic AI, RAG, and Intelligent Automation** into a single research assistance platform.

- **Multi-Agent Intelligence –** Multiple specialized agents collaborate to understand queries, retrieve information, analyze data, and generate research reports.
- **RAG-Based Accuracy** – Integrates Retrieval-Augmented Generation (RAG) with AI models to provide accurate, relevant, and up-to-date information.
- **Autonomous Research Workflow** – Agentic AI plans and executes research tasks with minimal human intervention, improving efficiency and productivity.
- **Intelligent Information Retrieval** – Collects relevant data from websites, documents, databases, and other trusted sources.
- **AI-Powered Summarization** – Converts large volumes of information into concise summaries and actionable insights.
- **Structured Report Generation** – Produces well-organized research reports with key findings, recommendations, and references.
- **IBM watsonx Orchestrate Integration** – Built using IBM watsonx Orchestrate to coordinate agents, automate workflows, and streamline research operations.
- **Scalable & Enterprise-Ready** – Designed on IBM Cloud technologies, ensuring reliability, scalability, security, and governance.

This makes the solution a **smart, autonomous, and efficient AI Research Assistant** capable of delivering high-quality research insights faster than traditional manual methods.

Git Hub Link

Please create public GitHub repository (Enable Add README button while creating repository) with format –

Please test and Paste the working GitHub URL –**https://github.com/boenipoojitha-design/Research-Agent.git**

Make sure you have Uploaded following three files into your GitHub repository

1) app.json file

2) yourproblemstatement.pdf file

3) your projectpresntation.pptx file

4)any other files

Thank You...